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Portofolio - Intensive Bootcamp

Introduction to Problem Solving

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Mini Task

- Halojek is a company with 2 main product:
 - Halofood : a service to provide food delivery
 - Haloride : a service to provide mobility using motor cycles
- The revenue on this quarter (Q3, 23) is decreased for 50% compared to same quarter last year (Q3, 22)

The cost for this quarter is as follow:

- Marketing cost : IDR 800 Bio,
- Labor cost: IDR 750 Bio
- Infrastructure cost: IDR 3,500 Bio

The cost for last quarter is as follow:

- Marketing cost : IDR 600 Bio,
- Labor cost: IDR 600 Bio
- Infrastructure cost: IDR 2000 Bio

Objective: Develop problem solving framework for the case above, you are a data analyst to deep dive the root cause of cost increasing on this quarter and also provide the alternative

Step 1: Clarify the Problem

Clue: Quantified the gap

Ideal Condition:

- Revenue ↑ / stable (0% decline)
- Cost Efficient (0% unnecessary increase)

Existing Condition:

- Revenue -50% (decrease)
- Cost + 57.8%

Gap

- Revenue underperformed by 50%
- Cost exceeded ideal efficiency by 57.8%
- Resulting in significant profitability decline

Rumus Growth

$$\text{Growth} = \frac{\text{Current} - \text{Previous}}{\text{Previous}} \times 100\%$$

Cost Last quarter:
= 600+600+2000
= 3200

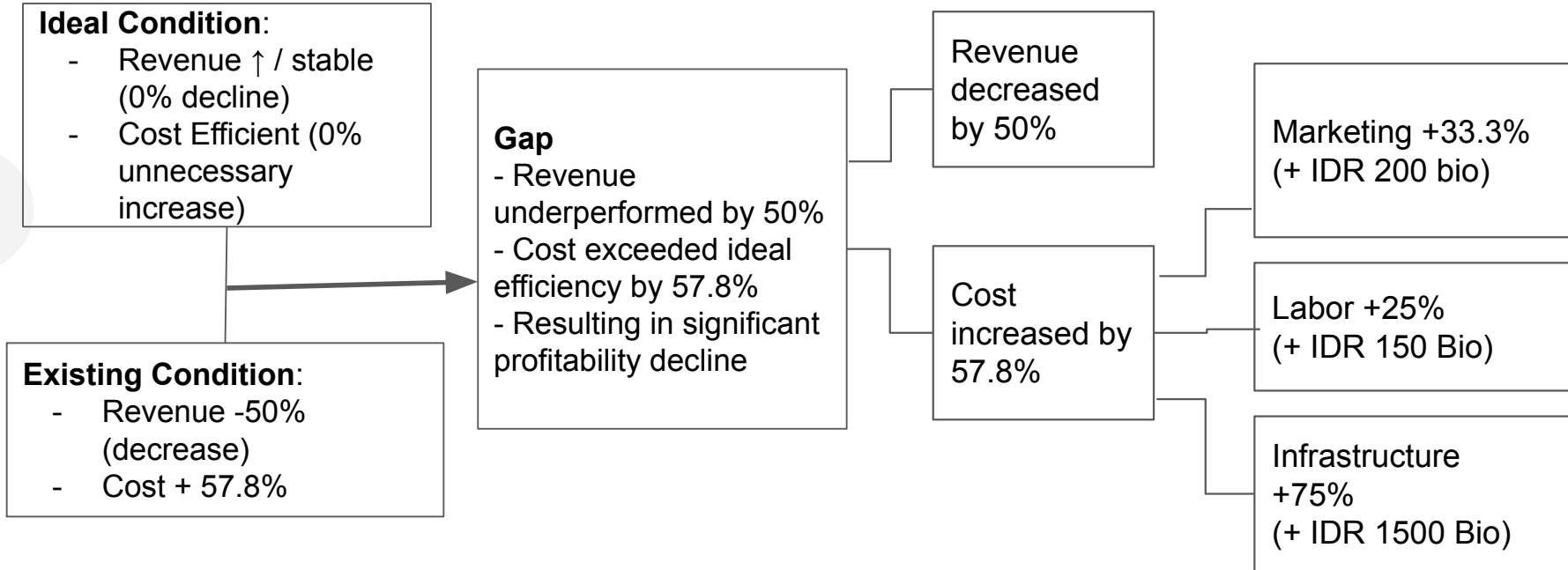
Cost Current quarter:
= 800+750+3500
= 5050

Selisih = 5050-3200 = 1850

Sub ke rumus growth:
= 1850 / 3200 x 100%
= 57.8% (Cost Increase)

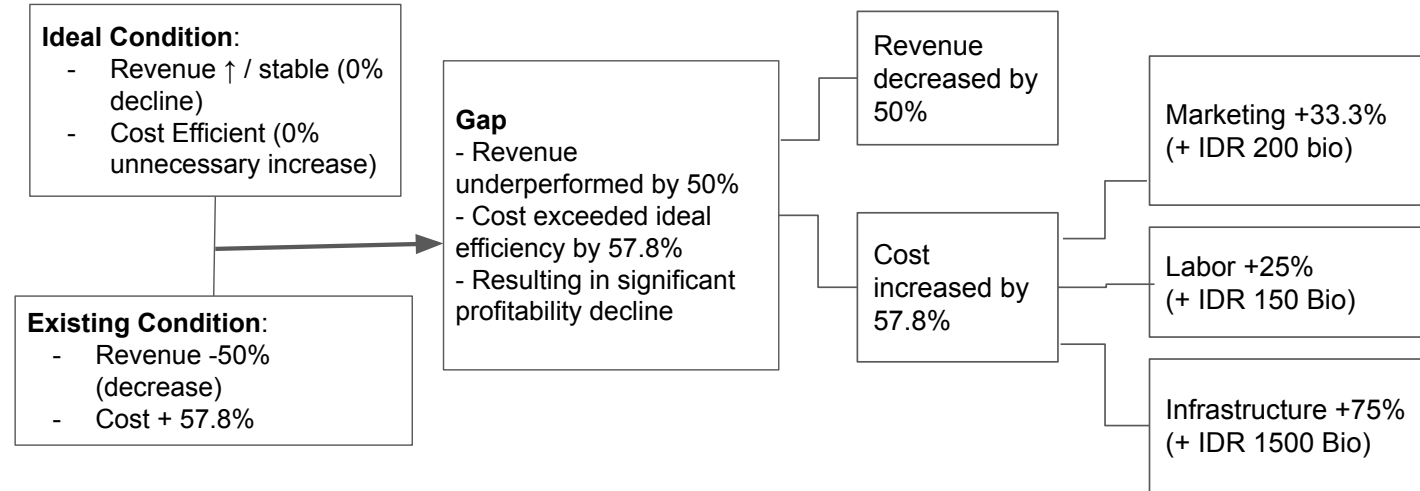
Step 2: Breakdown the Problem

Clue: Should be data driven | 4W - What, Where, Who, When



Step 3: Set the Objective

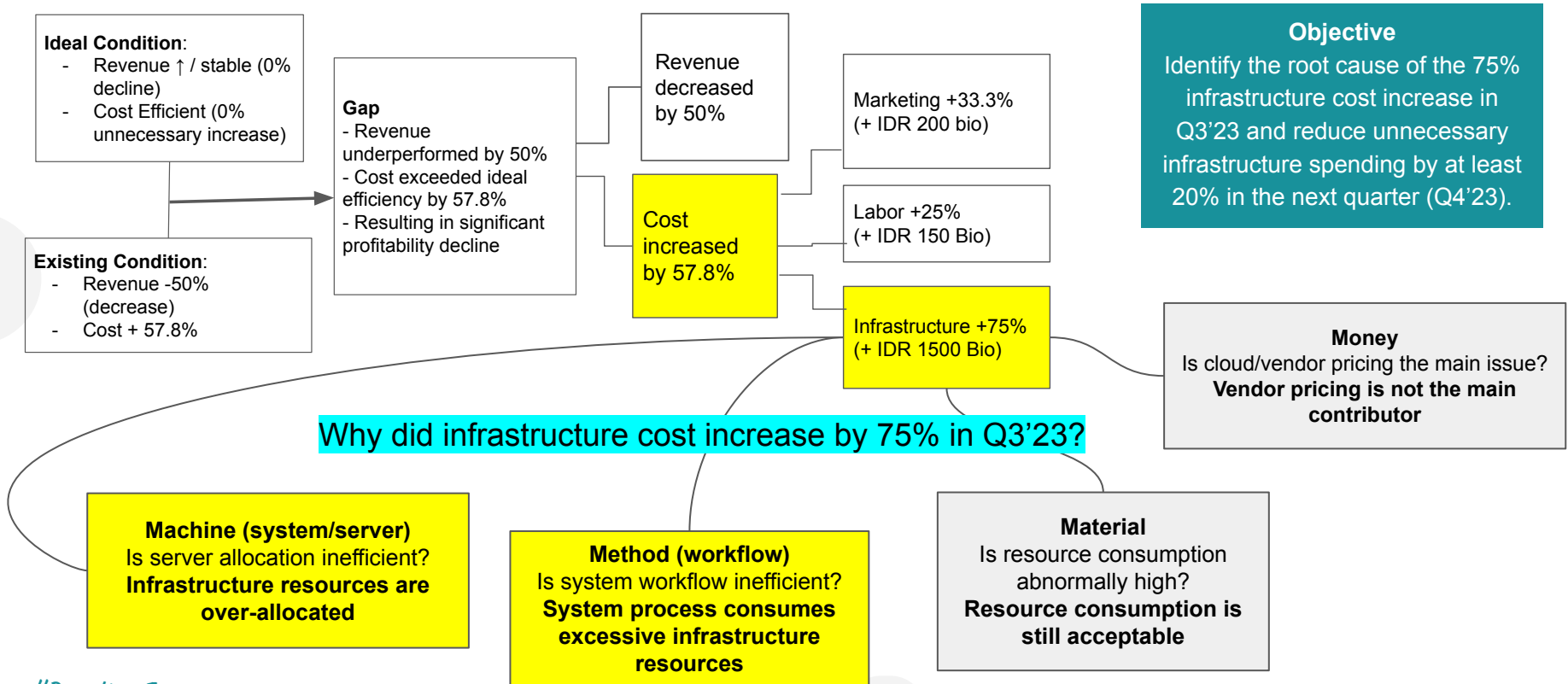
Clue: objective should be quantified and time-bound



Objective

Identify the root cause of the 75% infrastructure cost increase in Q3'23 and reduce unnecessary infrastructure spending by at least 20% in the next quarter (Q4'23).

Step 4: Define the root cause



5. Develop Countermeasures



Machine (system/server)
Server resources are over-allocated compared to actual demand

Problem	Alternative Countermeasure	Complexity	Cost	Speed	Conclusion
Over-allocated infrastructure resources (Machine)	Implement auto-scaling to match traffic demand	✓	✓	✓	✓ Recommended
	Manually reduce idle server capacity	✓	✓	✗	Short-term only
	Consolidate workloads into fewer servers	✗	✓	✗	Long-term option
Excessive infrastructure usage due to inefficient system process (Method)	Optimize API calls and eliminate redundant requests	✓	✓	✓	✓ Recommended
	Improve caching mechanism to reduce repeated processing	✓	✓	✓	✓ Recommended
	Optimize database queries and indexing	✗	✓	✗	Medium-term option

Method (workflow)
System architecture/process inefficiency causes excessive infra usage